

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2023 - 2024)

**JEE(Advanced)****FULL SYLLABUS****27-04-2024****JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)**

Time : 3 Hours

**PAPER-2 (OPTIONAL)**

Maximum Marks : 198

**READ THE INSTRUCTIONS CAREFULLY****GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

**OPTICAL RESPONSE SHEET :**

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

**DARKENING THE BUBBLES ON THE ORS :**

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take  **$g = 10 \text{ m/s}^2$**  unless otherwise stated.

**QUESTION PAPER FORMAT :**

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

**SOME USEFUL CONSTANTS**

**Atomic No.** : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

**Atomic masses** : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- |                                    |                                                              |
|------------------------------------|--------------------------------------------------------------|
| • Boltzmann constant               | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$                   |
| • Coulomb's law constant           | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$                   |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$  |
| • Speed of light in vacuum         | $c = 3 \times 10^8 \text{ ms}^{-1}$                          |
| • Stefan-Boltzmann constant        | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$                        |
| • Permeability of vacuum           | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$                |
| • Permittivity of vacuum           | $\epsilon_0 = \frac{1}{\mu_0 c^2}$                           |
| • Planck constant                  | $h = 6.63 \times 10^{-34} \text{ J-s}$                       |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

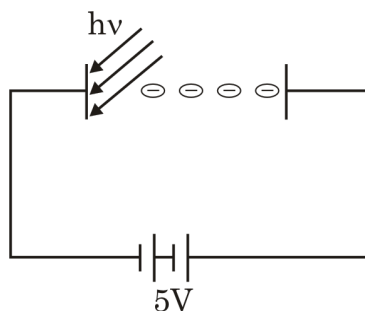
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

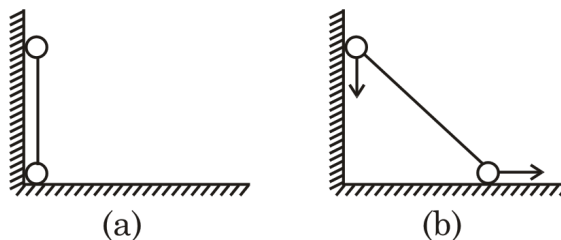
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Photons of energy 5 eV are incident on cathode. Electrons reaching the anode have kinetic energies varying from 6eV to 8eV.

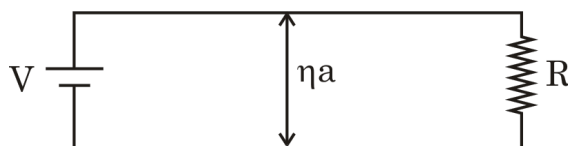


- (A) Work function of the metal is 2 eV.
- (B) Work function of the metal is 3 eV.
- (C) Current in the circuit is equal to saturation value.
- (D) Current in the circuit is less than saturation value.

2. We have a massless rod on which two small equally massive balls are attached as in figure (a). The floor and wall are smooth. When the system is released the balls begin to slide as shown in (b). Which of the following quantities are zero when upper ball just loses contact with the wall ?



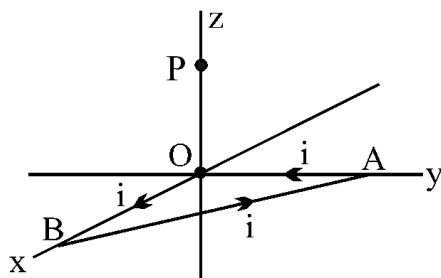
- (A) Tensile force in the rod. (B) Normal between wall & the ball.  
(C) Acceleration of the lower ball. (D) Acceleration of the upper ball.
3. Two long parallel wires of negligible resistance are connected at one end to a resistance  $R$  and at the other end to a ideal d.c. voltage source of emf  $V$ . The distance between the axes of the wires is  $\eta$  ( $\eta \gg 1$ ) times the cross-sectional radius of each wire. The upper wire gets positively charged with linear charge density  $\lambda$  and lower wire with linear charge density  $-\lambda$ . Choose the correct statement(s).



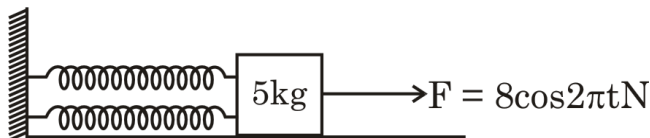
- (A) magnitude of linear charge density on wires =  $\frac{\pi \epsilon_0 V}{\ln \eta}$ .  
(B) magnitude of linear charge density on wires =  $\frac{2\pi \epsilon_0 V}{\ln \eta}$ .  
(C) for  $R = \sqrt{\frac{\mu_0}{\epsilon_0}} \frac{\ln \eta}{\pi}$ , the resultant force of interaction between the wires becomes zero.  
(D) for  $R = \sqrt{\frac{\mu_0}{\epsilon_0}} \frac{\ln \eta}{\pi}$ , the resultant force of interaction between the wires becomes zero.



4. AOB is a right angled isosceles triangle with sides  $OA = OB = \sqrt{2} a$  in xy plane. It is carrying current  $i$  as shown. The magnetic induction at the point P (0, 0, a)

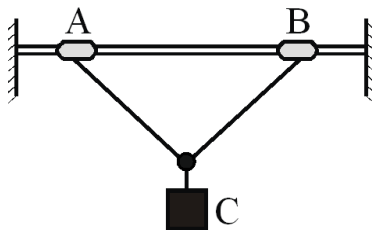


- (A) due to current in AOB is  $\vec{B} = -\frac{\mu_0 i}{2\sqrt{6} \pi a}(\hat{i} + \hat{j})$
- (B) due to current in BA is  $\vec{B} = \frac{\mu_0 i}{4\sqrt{6} \pi a}(\hat{i} + \hat{j} + \sqrt{3} \hat{k})$
- (C) due to current in AOB is  $\vec{B} = -\frac{\mu_0 i}{2\sqrt{3} \pi a}(\hat{i} + \hat{j})$
- (D) due to current in BA is  $\vec{B} = \frac{\mu_0 i}{4\sqrt{6} \pi a}(\hat{i} + \hat{j} + \sqrt{2} \hat{k})$
5. A 5kg block is attached to two springs of force constant 100 N/m each. The block experiences viscous friction. Select the correct statements for its oscillation in steady state. Take  $\pi^2 = 10$ ,  $b = 4$  kg/s.



- (A) The velocity amplitude is more than velocity amplitude for any other frequency of applied force.
- (B) The displacement amplitude is more than displacement amplitude for any other frequency of applied force.
- (C) Velocity amplitude is 2 m/s
- (D) Displacement amplitude is  $\frac{2}{\pi} \sqrt{\pi^2 - 1} \text{ m}$

6. Two identical sleeves A and B each of mass  $m$  can slide without friction on a fixed horizontal rod. A load C of mass  $2m$  is suspended from the mid-point of a light inextensible cord of length  $\ell = 3\sqrt{3}$  m connecting the sleeves A and B. Initially the sleeves are at a separation  $\ell$  and the load is held motionless. What can you conclude for subsequent motion after the load is released? Acceleration of free fall is  $10 \text{ m/s}^2$ .



- (A) Maximum speed of the sleeves is  $\sqrt{10}$  m/s.  
 (B) Maximum speed acquired by the load C is  $2\sqrt{5}$  m/s.  
 (C) The load descends a height 1.5 m to acquire maximum speed.  
 (D) Before the sleeves collide, the net force on the system consisting of the sleeves, the cord and the load becomes zero once.

SECTION-I (ii) : (Maximum Marks: 18)

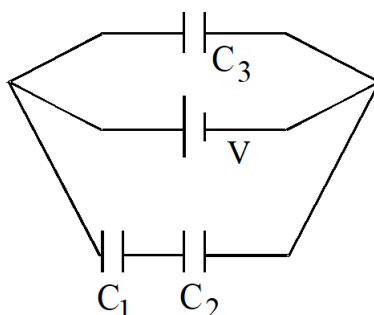
- This section contains **SIX (06)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +3 If ONLY the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered)

*Negative Marks* : -1 In all other cases

7. In the circuit shown in figure plates of capacitor  $C_2$  are gradually pulled apart. Then match the following :-



|     | List-I                                               |     | List-II          |
|-----|------------------------------------------------------|-----|------------------|
| (P) | Potential difference across $C_1$                    | (1) | Will remain same |
| (Q) | Energy stored in $C_3$                               | (2) | Will increase    |
| (R) | Potential difference across $C_2$                    | (3) | Will decrease    |
| (S) | Electric field between the plates of capacitor $C_1$ | (4) | None of these    |

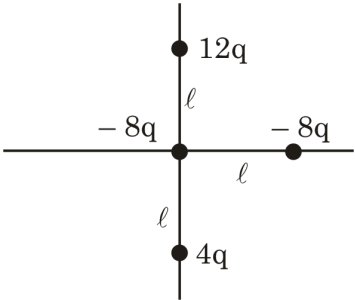
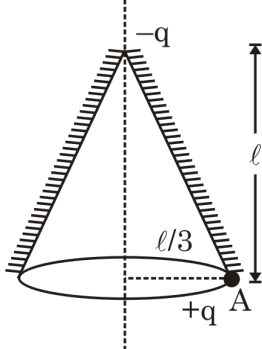
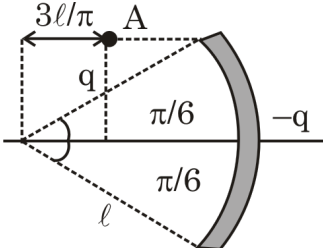
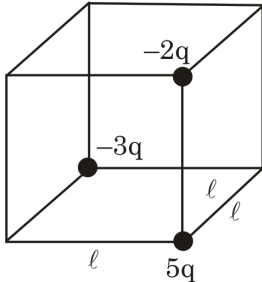
(A)  $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(B)  $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

(C)  $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

(D)  $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

8. List-I shows different charge distributions. List-II gives corresponding electric dipole moments.

|     | List-I                                                                                                                                                                                                                                                                   |     | List-II                   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------|
| (P) |  <p>Four point charges are kept as shown</p>                                                                                                                                            | (1) | $\sqrt{22}q\ell$          |
| (Q) |  <p>A uniformly charged hollow cone having height <math>\ell</math> and radius <math>\ell/3</math> carries charge <math>-q</math>. A charge <math>+q</math> is placed at point 'A'.</p> | (2) | $\frac{q\ell\sqrt{2}}{3}$ |
| (R) |  <p>Uniformly charged arc has charge <math>-q</math>. Another charge <math>q</math> is placed at point 'A'. Radius of arc is <math>\ell</math></p>                                    | (3) | $\frac{q\ell}{2}$         |
| (S) |  <p>A cube of side <math>\ell</math> has three charges as shown</p>                                                                                                                   | (4) | $8\sqrt{2}q\ell$          |

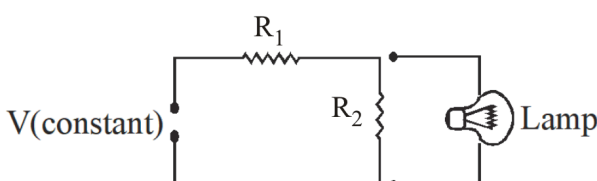
(A)  $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

(B)  $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 4$

(C)  $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

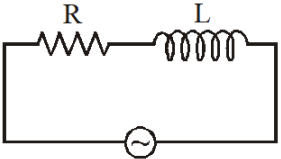
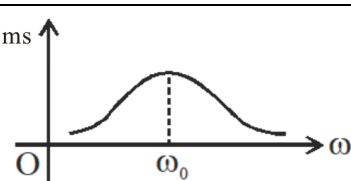
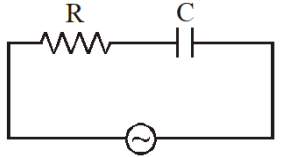
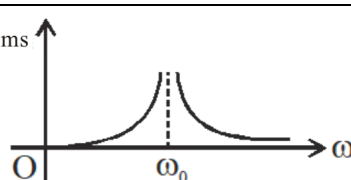
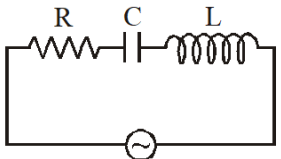
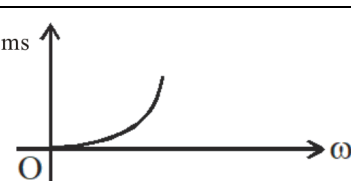
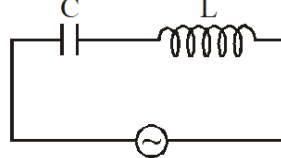
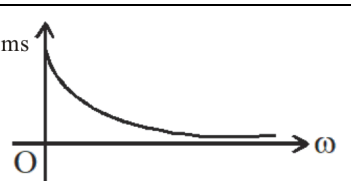
(D)  $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$

9. Match the entries in List-I with all the entries in List-II.

|     | List-I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     | List-II       |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------|
| (P) | <p>The current through <math>R_1</math> increase by 0.1 A on adding the lamp as shown below.</p>  <p>If <math>R_1/R_2 = 4</math>, what is the current through lamp in ampere?</p>                                                                                                                                                                                                                                              | (1) | 0             |
| (Q) | <p>A uniform thin bar of mass <math>M</math> is sliding along ice while rotating. The velocity of the bar's center of mass is <math>v</math> and its angular velocity is <math>\omega</math>. At the moment when the bar is perpendicular to the velocity of center of mass, the bar hits a stationary puck of mass <math>m</math> at a point on the end. The collision is elastic and as a result of collision, bar stays in place and puck slides away.</p> <p>What is the ratio <math>\frac{M}{m}</math>?</p> | (2) | $\frac{1}{2}$ |
| (R) | <p>In a meter bridge experiment, when the known resistor <math>R</math> and unknown resistor <math>X</math> are interchanged, the null point shifts by 60 cm. The ratio <math>\frac{R}{X}</math> could be</p>                                                                                                                                                                                                                                                                                                    | (3) | 1             |
| (S) | <p>A long straight wire carrying current <math>i</math> lies parallel to <math>z</math>-axis, along the line <math>y = 0, x = -a</math>. If <math>\vec{B}</math> be the magnetic field at any point on <math>y</math>-axis, <math>\int_{-\infty}^{\infty} \vec{B} \cdot d\vec{y} = \mu_0 ni</math>, find <math>n</math>.</p>                                                                                                                                                                                     | (4) | 4             |

- (A)  $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$
- (B)  $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 4$
- (C)  $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$
- (D)  $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 4; S \rightarrow 3$

10. A.C. circuits are shown in list-I and  $i_{\text{rms}}$  vs  $\omega$  graph is shown in list-II match the corresponding graph.

|     | List-I                                                                             |     | List-II                                                                             |
|-----|------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------|
| (P) |   | (1) |   |
| (Q) |   | (2) |   |
| (R) |   | (3) |   |
| (S) |  | (4) |  |

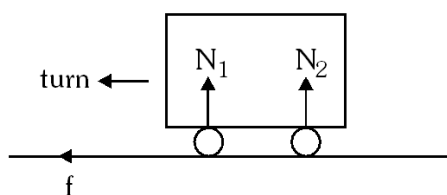
(A)  $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$

(B)  $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

(C)  $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(D)  $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

11. A car is taking a turn on a rough horizontal road without slipping as shown in figure. Let  $F$  is centripetal force,  $f$  the force of friction,  $N_1$  and  $N_2$  are two normal reactions. As the speed of car is increased match the following two Lists.



|     | List-I |     | List-II                |
|-----|--------|-----|------------------------|
| (P) | $N_1$  | (1) | Will increase          |
| (Q) | $N_2$  | (2) | Will decrease          |
| (R) | $F/f$  | (3) | Will remains unchanged |
| (S) | $F$    | (4) | Cannot say anything    |

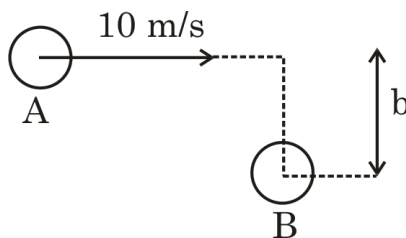
(A)  $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

(B)  $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 1$

(C)  $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 4$

(D)  $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

12. A ball of mass 1 kg is projected towards another stationary ball of mass 1 kg. Both balls have same radius of 10 cm. If there is a collision, the collision is elastic. List-I gives the value of impact parameter and List-II gives velocity (magnitude) of ball B after collision. Match them.



|     | List-I      |     | List-II |
|-----|-------------|-----|---------|
| (P) | $b = 0$     | (1) | 10 m/s  |
| (Q) | $b = 30$ cm | (2) | 0 m/s   |
| (R) | $b = 12$ cm | (3) | 8 m/s   |
| (S) | $b = 16$ cm | (4) | 6 m/s   |

- (A)  $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$   
 (B)  $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$   
 (C)  $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$   
 (D)  $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

SECTION-II : (Maximum Marks: 24)

**MATCHING LIST TYPE INTEGER QUESTION (ANSWER AS FOUR DIGIT INTEGER)**

- This section contains **SIX (06)** questions.
- The answer to each question is a four digit integer ranging from 1111 to 4444, both inclusive
- Each question having two matching lists. Choices for the correct combination of options form List-I (P), (Q), (R), (S) and List-II (1), (2), (3), (4)
- Given matching type questions in which List-I (PQRS) match with List-II (1, 2, 3, 4) and represent your answer as four digit integer for example correct match for  $P \rightarrow 2$ ,  $Q \rightarrow 3$ ,  $R \rightarrow 1$ ,  $S \rightarrow 4$ . Then you have fill the answer in Section-II as 2314.
- Each matching in List-II may be/may not be correct & can be used more than one time for matching in List-I. For example your answer can be 2233, 1333, 1433, 4321 etc...
- For each question, marks will be awarded in one of the following categories :

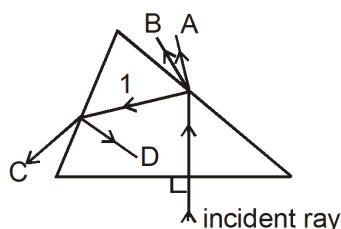
**Full Marks** : +4 If ONLY the correct numerical value is entered.

**Zero Marks** : 0 In all other cases.

1. In the following question x (an unstable nucleus) decays to another nucleus y. In List-I different type of decay processes are mentioned. Atomic mass of element of nucleus x and element of nucleus y are  $M_x$  and  $M_y$  respectively. The atomic mass of He is  $M_{He}$ . The mass of electron is  $m_e$ . Now match the entries of List-I with List-II.

|     | List-I (Decay process)                               |     | List-II (Mass defect) |
|-----|------------------------------------------------------|-----|-----------------------|
| (P) | $x \rightarrow y + \alpha$                           | (1) | $M_x - M_y - m_e$     |
| (Q) | $x \rightarrow y + {}_{-1}e^0$                       | (2) | $M_x - M_y - M_{He}$  |
| (R) | $x \rightarrow y + {}_{+1}e^0$                       | (3) | $M_x - M_y - 2m_e$    |
| (S) | $x + {}_{-1}e^0 \rightarrow y + \nu + \text{X-rays}$ | (4) | $M_x - M_y$           |

2. A white light ray is incident on a glass prism, and it create four refracted rays A, B, C and D. Match the refracted rays with the colours given (1 & D are rays due to total internal reflection.):



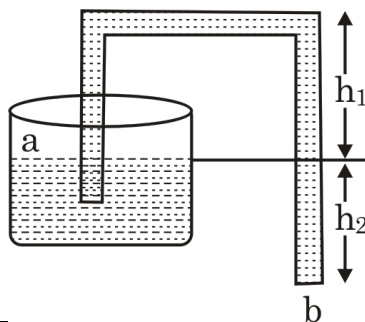
|     | List-I (Ray) |     | List-II (Colour) |
|-----|--------------|-----|------------------|
| (P) | A            | (1) | Red              |
| (Q) | B            | (2) | Green            |
| (R) | C            | (3) | Yellow           |
| (S) | D            | (4) | Blue             |



3. A polarized electromagnetic plane beam  $\vec{E} = 2\sqrt{\pi} \left( 3\hat{i} + 4\hat{j} \right) \text{ N/C} \sin \left( \omega \left( t + \frac{Z}{C} \right) \right)$  is travelling along  $-ve$   $z$  direction. A polaroid is placed in its path at origin so that polaroid is in  $xy$  plane. List-I indicates pass axis & List-II emergent intensity. Indicated in List-I & List-II respectively. Match them

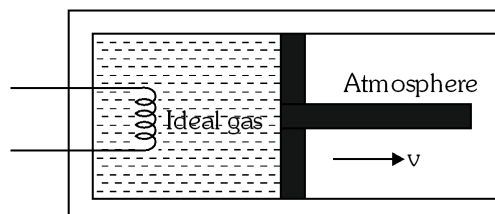
|     | List-I<br>(Pass axis along) |     | List-II<br>(Intensity) |
|-----|-----------------------------|-----|------------------------|
| (P) | $x = y$                     | (1) | $\frac{49}{120} w/m^2$ |
| (Q) | $x = 0$                     | (2) | $\frac{48}{125} w/m^2$ |
| (R) | $y = 0$                     | (3) | $\frac{3}{20} w/m^2$   |
| (S) | $3x = 4y$                   | (4) | $\frac{4}{15} w/m^2$   |

4. A siphon transfers liquid from high level to low level as shown in figure. If  $\rho$  is density of liquid,  $P_0$  atmospheric pressure,  $V_a$  is velocity at point a (on the surface of water in the tank) and  $V_b$  is velocity at point b (at the right open end of the siphon), then ( $V_b \neq 0$ ).  $P_a$ ,  $P_b$  are pressures at points a and b respectively.



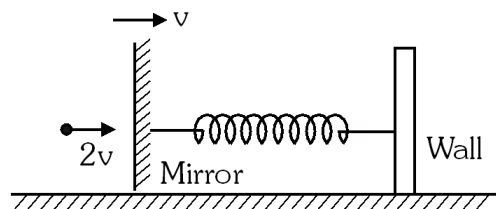
|     | List-I                             |     | List-II              |
|-----|------------------------------------|-----|----------------------|
| (P) | $(h_1 + h_2) - \frac{P_0}{\rho g}$ | (1) | is equal to one      |
| (Q) | $V_a - V_b$                        | (2) | is less than zero    |
| (R) | $h_1$                              | (3) | is greater than zero |
| (S) | $\frac{P_b - P_a}{\rho g h_1}$     | (4) | is equal to zero     |

5. An ideal gas at initial pressure  $P_0$  (= atmospheric pressure) and volume  $V_0$  is enclosed by a light piston in a cylinder as shown. Initially the piston is in static equilibrium and the gas is in thermodynamic equilibrium. Now we start to move the piston outwards at a constant slow velocity  $v$ . The cross sectional area of piston is  $A$ . While the piston is moving we can deliver heat to the gas through a heating filament and pull the piston in such a way that temperature of gas remains constant. Apart from the heat transfer between the gas and the heating filament all other heat exchange can be neglected and the atmospheric pressure is constant. Mark the quantities in List-I with their appropriate variation in List-II (time  $t$  on x-axis)



|     | List-I                                                                             |     | List-II |
|-----|------------------------------------------------------------------------------------|-----|---------|
| (P) | Pressure $P$ of gas on y-axis                                                      | (1) |         |
| (Q) | Volume $V$ of gas on y-axis                                                        | (2) |         |
| (R) | Force applied by pulling agent (apart from atmosphere) on the piston on y-axis     | (3) |         |
| (S) | Amount of cumulative work done by gas on piston from start till time 't' on y-axis | (4) |         |

6. A plane mirror is tied to the free end of an ideal spring. The other end of the spring is attached to a wall. The spring with mirror is held vertically to the floor can slide along it smoothly. When the spring is at its natural length, the mirror is found to move at speed  $v$  with respect to ground. An object moving towards mirror with speed  $2v$  with respect to ground. Match the following : (Assume no collision before mirror comes to rest for 1<sup>st</sup> time)



|     | List-I                                                                               |     | List-II |
|-----|--------------------------------------------------------------------------------------|-----|---------|
| (P) | Speed of image with respect to ground when spring at natural length                  | (1) | $v$     |
| (Q) | Speed of image with respect to mirror when spring at natural length                  | (2) | 0       |
| (R) | Speed of image with respect to object when spring is at natural length               | (3) | $2v$    |
| (S) | Speed of image with respect to ground frame when spring at maximum compressed state. | (4) | $3v$    |

**PART-2 : CHEMISTRY**  
**SECTION-I (i) : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

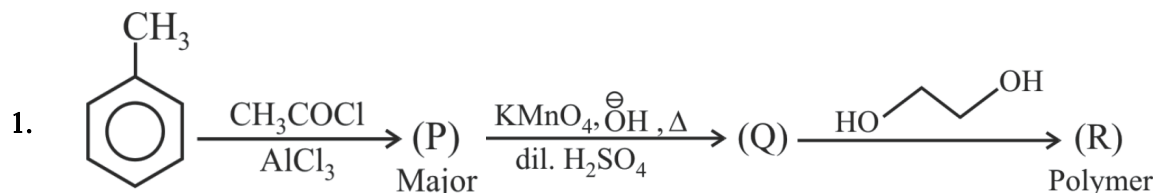
*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

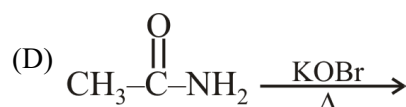
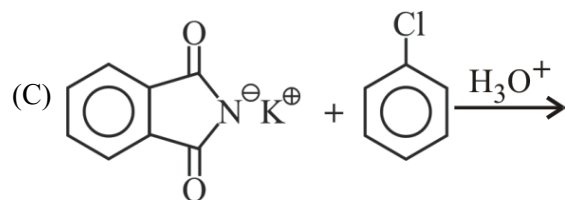
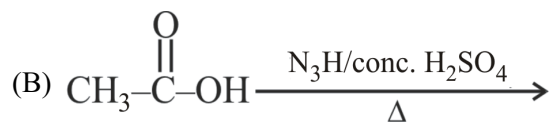
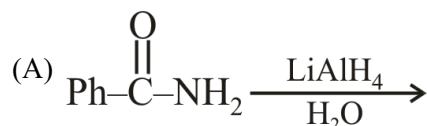
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.



Choose correct option(s) ?

- (A) (Q) is soluble in aq.  $\text{NaHCO}_3$  as well as in aq.  $\text{NaOH}$
- (B) Formation of (R) is step growth polymerisation
- (C) (P) will give yellow crystal with  $\text{NaOI}$
- (D) (R) is polymer terylene

2. Which of the reaction(s) will produce 1° amine?



3. Which of the following statement(s) are correct for nitrogen family elements ?

- (A) Oxidation state from +1 to +4 for nitrogen tend to disproportionate in acidic medium.
- (B) Phosphorous and arsenic can form  $d\pi-d\pi$  bond with transition metals.
- (C) The only basic oxide of N-family is  $\text{Bi}_2\text{O}_5$ .
- (D) The only trihalide  $\text{BiF}_3$  is ionic in nature.

4. Select the correct statement(s).

- (A) Compounds of Lanthanoids resemble with alkali metal in properties.
- (B) Atomic size of lanthanoids decreases regularly on moving left to right.
- (C) Lanthanoids show +3 oxidation state.
- (D) Lanthanoids are soft and silvery white.

5. Identify correct statement(s) :-

- (A) Of the three types of cubic lattices, SC, BCC & FCC those having greatest & least densities for identical atoms are FCC & SC respectively.
- (B) A plot between vapour pressure of a liquid & its absolute temperature results in a straight line.
- (C) In CCP three dimensional structure, possible distances between octahedral voids are  $\frac{a}{2}, \frac{a}{\sqrt{2}}$ .  
[a = Edge length of unit cell]
- (D) In FCC unit cell the possible distances between tetrahedral voids are  $\frac{a}{2}, \frac{a}{\sqrt{2}}, \frac{a\sqrt{3}}{2}$  (where a is edge length of unit cell).

6. According to the Arrhenius equation:

- (A) In Arrhenius equation, factor  $e^{-E_a/RT}$  corresponds to fraction of reactant molecules that have kinetic energy equal to activation energy.
- (B) Rate constant increases with increases in temperature. This is mainly due to a greater number of collisions whose energy exceeds the activation energy.
- (C) Higher the magnitude of activation energy of a reaction, stronger is the temperature dependence of its rate constant.
- (D) The pre-exponential factor depends on the frequency at which collisions between reactant molecules per unit volume per unit time occur, irrespective of their energy.

SECTION-I (ii) : (Maximum Marks: 18)

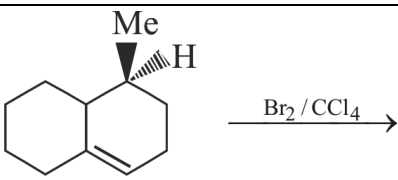
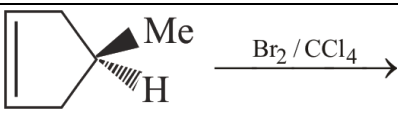
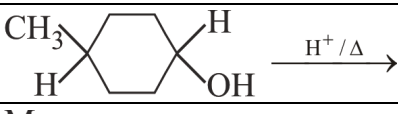
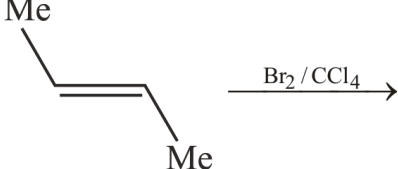
- This section contains **SIX (06)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +3 If ONLY the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered)

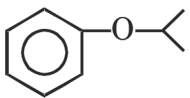
*Negative Marks* : -1 In all other cases

7. Matching List.

| List-I |                                                                                     | List-II<br>(About product mixture) |                    |
|--------|-------------------------------------------------------------------------------------|------------------------------------|--------------------|
| (P)    |    | (1)                                | Optically active   |
| (Q)    |   | (2)                                | Diastereoisomers   |
| (R)    |  | (3)                                | Optically inactive |
| (S)    |  | (4)                                | Meso               |

- (A) P → 1,2; Q → 3; R → 3; S → 3,4  
 (B) P → 1,3; Q → 2; R → 3; S → 3,4  
 (C) P → 1,2; Q → 3; R → 1; S → 2,3  
 (D) P → 1,2; Q → 4; R → 2; S → 3,4

8. Match the list

| List-I<br>(Reactions) |                                                                                                                                                                                                                                                                                                                                                          | List-II<br>(A and B is differentiated by) |                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------|
| (P)                   | $\text{Ph} - \text{CH} = \text{O} \xrightarrow[(2) \text{H}^+]{(1) \text{Conc. } \text{OH}^-} \text{A} + \text{B}$                                                                                                                                                                                                                                       | (1)                                       | $\text{FeCl}_3$  |
| (Q)                   | $\text{CH}_3 - \text{C} \equiv \text{CH} \xrightarrow[\text{dil. H}_2\text{SO}_4]{\text{Hg}^{+2}} \text{A}$<br>$\text{CH}_3 - \text{C} \equiv \text{CH} \xrightarrow[(2) \text{H}_2\text{O}_2, \text{HO}^-]{(1) \text{BH}_3, \text{THF}} \text{B}$                                                                                                       | (2)                                       | Fehling test     |
| (R)                   | $\text{Ph} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{OH} \xrightarrow{\Delta} (\text{A})$<br>$\text{CH}_3 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \underset{\text{CH}_3}{\text{CH}} - \overset{\text{O}}{\parallel} \text{C} - \text{OH} \xrightarrow{\Delta} (\text{B})$ | (3)                                       | $\text{NaHCO}_3$ |
| (S)                   |  $\xrightarrow{\text{H}^+/\text{H}_2\text{O}} \text{A} + \text{B}$                                                                                                                                                                                                      | (4)                                       | Iodoform test    |

(A)  $\text{P} \rightarrow 3; \text{Q} \rightarrow 2; \text{R} \rightarrow 4; \text{S} \rightarrow 2$

(B)  $\text{P} \rightarrow 2; \text{Q} \rightarrow 3; \text{R} \rightarrow 1; \text{S} \rightarrow 3$

(C)  $\text{P} \rightarrow 2; \text{Q} \rightarrow 3; \text{R} \rightarrow 3; \text{S} \rightarrow 1$

(D)  $\text{P} \rightarrow 3; \text{Q} \rightarrow 2; \text{R} \rightarrow 4; \text{S} \rightarrow 4$

9. Match list-I with list-II.

| List-I<br>[Property of a metal (M)] |                                                                                              | List-II<br>[Metal (M)] |    |
|-------------------------------------|----------------------------------------------------------------------------------------------|------------------------|----|
| (P)                                 | "M" produces $\text{H}_2$ with KOH solution                                                  | (1)                    | Au |
| (Q)                                 | "M" produces $\text{M}^{2+}$ ion when reacts with dil. HCl                                   | (2)                    | Zn |
| (R)                                 | "M" does not dissolve in dil. HCl but dissolves in dil. $\text{HNO}_3$                       | (3)                    | Hg |
| (S)                                 | "M" does not dissolve in conc. $\text{HNO}_3$ dil. $\text{HNO}_3$ or $\text{H}_2\text{SO}_4$ | (4)                    | Fe |

(A)  $\text{P} \rightarrow 1; \text{Q} \rightarrow 2; \text{R} \rightarrow 3; \text{S} \rightarrow 2$

(B)  $\text{P} \rightarrow 4; \text{Q} \rightarrow 1; \text{R} \rightarrow 2; \text{S} \rightarrow 3$

(C)  $\text{P} \rightarrow 2; \text{Q} \rightarrow 4; \text{R} \rightarrow 1; \text{S} \rightarrow 4$

(D)  $\text{P} \rightarrow 2; \text{Q} \rightarrow 2; \text{R} \rightarrow 4; \text{S} \rightarrow 3$



10. Match list-I with list-II.

| List-I |                                     | List-II |                                                         |
|--------|-------------------------------------|---------|---------------------------------------------------------|
| (P)    | $K_4[Fe(CN)_6] > K_3[Fe(CN)_6]$     | (1)     | Correct order of stability                              |
| (Q)    | $[Co(NH_3)_6]Cl_3 < [Co(en)_3]Cl_3$ | (2)     | Correct order of magnetic moment                        |
| (R)    | $[Co(NH_3)_6]Cl_3 < Na_3[CoF_6]$    | (3)     | Correct order of oxidation number of central metal atom |
| (S)    | $[NiF_6]^{2-} > [NiF_6]^{4-}$       | (4)     | Correct order of number of $t_{2g}$ electrons           |

(A)  $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 1,3$

(B)  $P \rightarrow 3; Q \rightarrow 3,4; R \rightarrow 1,3,4; S \rightarrow 3,4$

(C)  $P \rightarrow 2,3; Q \rightarrow 2,4; R \rightarrow 4; S \rightarrow 3,4$

(D)  $P \rightarrow 3,4; Q \rightarrow 2,4; R \rightarrow 4; S \rightarrow 2,3,4$

11.

| List-I |                                             | List-II |                                      |
|--------|---------------------------------------------|---------|--------------------------------------|
| (P)    | Orbital angular momentum of an electron     | (1)     | $\sqrt{s(s+1)} \frac{h}{2\pi}$       |
| (Q)    | Angular momentum of an electron in an orbit | (2)     | $\sqrt{n(n+2)}$                      |
| (R)    | Spin angular momentum of an electron        | (3)     | $\frac{nh}{2\pi}$                    |
| (S)    | Spin magnetic moment of atom/ion            | (4)     | $\sqrt{\ell(\ell+1)} \frac{h}{2\pi}$ |

(A)  $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(B)  $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(C)  $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$

(D)  $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

12. Match list-I with list-II. (Assuming all the solutes are non volatile and all solutions are ideal.)

| List - I (Experiment) |                                                                                        | List - II (Result) |                                        |
|-----------------------|----------------------------------------------------------------------------------------|--------------------|----------------------------------------|
| (P)                   | 10 milli moles of $HgI_2$ are added to 10 ml of 2 M aqueous NaI                        | (1)                | Osmotic pressure of solution increases |
| (Q)                   | 10 milli moles of KCl are added to 10 ml of 1 M aqueous NaCl                           | (2)                | Vapour pressure of solution increases  |
| (R)                   | 10 ml 0.1 M NaOH aqueous solution is added to 10 ml 0.1 M HCl aqueous solution         | (3)                | Boiling point of solution increases    |
| (S)                   | 10 ml of 0.1 NaOH aqueous solution is added to 10 ml 0.1 M $CH_3COOH$ aqueous solution | (4)                | Freezing point of solution increases   |

(A)  $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 4$

(B)  $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$

(C)  $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 2$

(D)  $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 4$

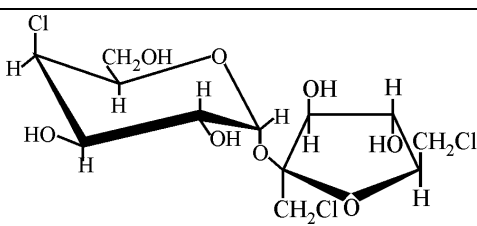
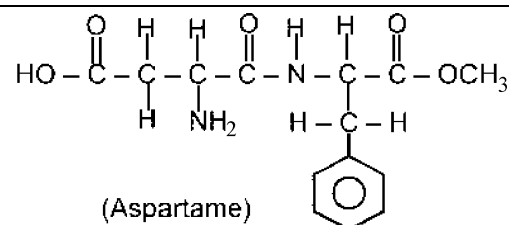
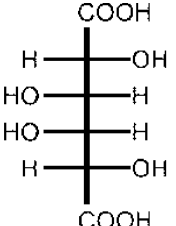
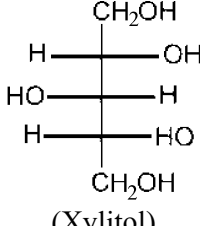
SECTION-II : (Maximum Marks: 24)

**MATCHING LIST TYPE INTEGER QUESTION (ANSWER AS FOUR DIGIT INTEGER)**

- This section contains **SIX (06)** questions.
- The answer to each question is a four digit integer ranging from 1111 to 4444, both inclusive
- Each question having two matching lists. Choices for the correct combination of options form List-I (P), (Q), (R), (S) and List-II (1), (2), (3), (4)
- Given matching type questions in which List-I (PQRS) match with List-II (1, 2, 3, 4) and represent your answer as four digit integer for example correct match for  $P \rightarrow 2$ ,  $Q \rightarrow 3$ ,  $R \rightarrow 1$ ,  $S \rightarrow 4$ . Then you have fill the answer in Section-II as 2314.
- Each matching in List-II may be/may not be correct & can be used more than one time for matching in List-I. For example your answer can be 2233, 1333, 1433, 4321 etc...
- For each question, marks will be awarded in one of the following categories :

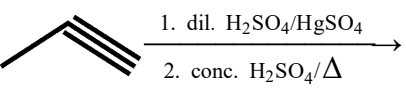
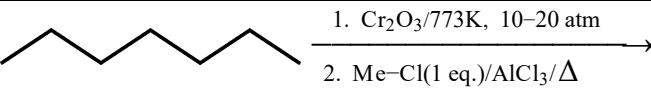
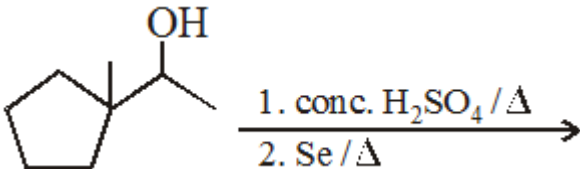
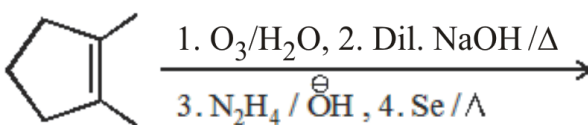
**Full Marks** : +4 If ONLY the correct numerical value is entered.

**Zero Marks** : 0 In all other cases.

| 1.  | List-I<br>(Compound)                                                                                                 | List-II<br>(Characteristic)             |
|-----|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| (P) |  <p>(Sucralose)</p>                | (1) A derivative of dipeptide           |
| (Q) |  <p>(Aspartame)</p>               | (2) A derivative of disaccharide        |
| (R) |  <p>(Galactaric Aldaric acid)</p> | (3) Reduction product of an aldopentose |
| (S) |  <p>(Xylitol)</p>                 | (4) Oxidation product of aldohexose     |

Match each process given in **List-I** with one or more effect(s) in **List-II**.

2.

| List-I<br>(Reactant) |                                                                                   | List-II<br>(Product) |                         |
|----------------------|-----------------------------------------------------------------------------------|----------------------|-------------------------|
| (P)                  |  | (1)                  | 1,3-Dimethyl benzene    |
| (Q)                  |  | (2)                  | Toluene                 |
| (R)                  |  | (3)                  | 1,2-dimethyl benzene    |
| (S)                  |  | (4)                  | 1,3,5-Trimethyl benzene |

Match each process given in **List-I** with one or more effect(s) in **List-II**.

3.

| List-I<br>(Name of mineral) |              | List-II<br>(Metal present in given mineral) |    |
|-----------------------------|--------------|---------------------------------------------|----|
| (P)                         | Fluorapatite | (1)                                         | Zn |
| (Q)                         | Kaolinite    | (2)                                         | Al |
| (R)                         | Magnesite    | (3)                                         | Ca |
| (S)                         | Calamine     | (4)                                         | Mg |

4.

| List-I<br>(Reaction) |                                                                               | List-II<br>(colour of metal containing product ) |            |
|----------------------|-------------------------------------------------------------------------------|--------------------------------------------------|------------|
| (P)                  | $\text{MnO}_2 + \text{KOH} \xrightarrow{\text{O}_2}$                          | (1)                                              | Deep blue  |
| (Q)                  | $\text{Co}^{2+} + \text{KNO}_2 \xrightarrow{\text{H}^+}$                      | (2)                                              | Yellow     |
| (R)                  | $\text{Mn}^{+2} + \text{S}_2\text{O}_8^{2-} + \text{H}_2\text{O} \rightarrow$ | (3)                                              | Dark green |
| (S)                  | $\text{Ni}(\text{OH})_2 + \text{NH}_4^+ + \text{OH}^- \rightarrow$            | (4)                                              | Purple     |

5.

| List-I (For a sample of ideal gas) |                                                | List-II |                            |
|------------------------------------|------------------------------------------------|---------|----------------------------|
| (P)                                | $\Delta S_{\text{Sys}}$ for isobaric process   | (1)     | $nC_V \ln \frac{T_2}{T_1}$ |
| (Q)                                | $\Delta S_{\text{Sys}}$ for isothermal process | (2)     | $nRT \ln \frac{V_1}{V_2}$  |
| (R)                                | Work done in reversible isothermal process     | (3)     | $-nR \ln \frac{V_1}{V_2}$  |
| (S)                                | $\Delta S_{\text{Sys}}$ for isochoric process  | (4)     | $nC_P \ln \frac{T_2}{T_1}$ |

6. Match the Lists.

| List-I |                                                                                                                                                                                             | List-II (pH) |     |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----|
| (P)    | 2.5 ml $\frac{2}{5}$ M weak base ( $K_b = 10^{-10}$ ) is mixed with $\frac{2}{15}$ M HCl for complete neutralisation                                                                        | (1)          | 5   |
| (Q)    | 1 mole $\text{Na}_3\text{A}$ is mixed with 2.5 mole of HCl to give 2 L solution ( $\text{pK}_{a1}$ , $\text{pK}_{a2}$ , $\text{pK}_{a3}$ of $\text{H}_3\text{A}$ are 3, 6, 10 respectively) | (2)          | 2.5 |
| (R)    | pH at which 0.1 M $\text{CH}_3\text{COOH}$ dissociates $\frac{1000}{11}\%$ ( $\text{pK}_a$ of $\text{CH}_3\text{COOH} = 5$ )                                                                | (3)          | 3   |
| (S)    | 1 L 0.1 M HA is mixed with 99 L water ( $\text{pK}_a$ of HA = $\text{pK}_b$ of $\text{A}^-$ )                                                                                               | (4)          | 6   |

**PART-3 : MATHEMATICS**  
**SECTION-I (i) : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

- Let  $f(x) = \lim_{n \rightarrow \infty} (-n^3) \left( \left| 2 \tan^{-1} (e^{|x|} - e) - \frac{1}{n^3} \right| - 2 \left| \tan^{-1} (e^{|x|} - e) \right| \right)$ ,  $x \in \mathbb{R}$ , then which of the following statement(s) is(are) correct ?
  - The number of points where  $y = f(x)$  is discontinuous is 1
  - The number of points where  $y = |f(x)|$  is discontinuous is 1
  - $f\left(\frac{1}{2}\right) + f(3) = 0$
  - Number of solution of  $f(x) = \text{sgn}(x)$  is infinite. (Where  $\text{sgn}(x)$  is the signum function of  $x$ )
- If  $f : \mathbb{R} \rightarrow \mathbb{R}$  be differentiable function such that  $(f(x))^7 = x - f(x)$ , then
  - $\int_0^{\sqrt{2}} (f^{-1}(x)) dx = 3$
  - $f(x)$  is increasing  $\forall x \in \mathbb{R}$
  - $\frac{d}{dx} (f^{-1}(x))$  at  $x = 2$  is 499
  - Let  $g(x)$  be the inverse of  $f(x)$ . Then  $g''(1) = 42$

3. P denotes plane containing all points equidistant from A(-4, 2, 1), B(2, -4, 3). Q be the plane  $x - y + cz = 1$ , where  $c \in \mathbb{R}$ , then

(A) If the line L with equation  $\frac{x-1}{1} = \frac{y+2}{3} = \frac{z-7}{-1}$  intersect the plane P at  $(x_0, y_0, z_0)$  then  $x_0 + y_0 + z_0 = 12$

(B) If angle between the planes P and Q is  $45^\circ$  then product of all values of c is -2

(C) The plane P is parallel to plane Q iff  $c = \frac{1}{3}$

(D) The plane P is parallel to plane Q iff  $c = 1$

4. In a drawer, Mr. Abhishek has 5 pairs of socks, each pair having a different colour. On Monday, Mr. Abhishek selects two socks at random from the 10 socks in the drawer. On Tuesday, Mr. Abhishek selects 2 of the remaining 8 socks at random; and on Wednesday, two of the remaining 6 socks at random. The probability that Wednesday is the first day Mr. Abhishek select same colour socks is  $\frac{m}{n}$ , where m and n are relatively prime positive integers, then

(A)  $m + n = 341$

(B)  $\lim_{x \rightarrow 0^+} x \left( \left\lfloor \frac{1}{x} \right\rfloor + \left\lfloor \frac{2}{x} \right\rfloor + \left\lfloor \frac{3}{x} \right\rfloor + \dots + \left\lfloor \frac{m}{x} \right\rfloor \right) = m + n$  ([.] is greatest integer function)

(C) Number of ordered pairs (x, y), (x and y are natural numbers), satisfying the condition  $x + y < 28$  and  $x + y \geq 6$ , is  $m + n$

(D)  $\lim_{x \rightarrow 0} x \left( \left\lfloor \frac{1}{x} \right\rfloor + \left\lfloor \frac{2}{x} \right\rfloor + \left\lfloor \frac{3}{x} \right\rfloor + \dots + \left\lfloor \frac{m}{x} \right\rfloor \right) = n - m$  ([.] is greatest integer function)

5. A line divides a plane into 2 regions. Two lines divide the plane into maximum 4 regions. If  $L_n$  is the maximum number of regions divided by n lines then which of the following is/are true :

(A)  $L_{20} = 211$

(B)  $L_{10} = 56$

(C)  $L_{15} = 121$

(D)  $L_{25} = 326$

6. If  $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ , then which of the following is(are) INCORRECT (where  $A^{-1}$  is inverse of matrix A,

I is an unit matrix of order 3, adj A is adjoint of matrix A,  $|A|$  denotes determinant value of matrix A.)

(A)  $\frac{1}{81} |\text{adj}(2A) \cdot \text{adj}(3A^{-1}) \cdot A^{-1}|$  is equal to 288

(B)  $(A^{-1})^2 = \frac{1}{2}(A - 5I + 7A^{-1})$

(C)  $|A^5 - 2A^4|$  is equal to 16

(D)  $A^{-1} = \frac{1}{2}(A^2 - 5A + 6I)$

SECTION-I (ii) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +3 If ONLY the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered)

*Negative Marks* : -1 In all other cases

7. Match the following :

|     | List-I                                                                                                                                                                                                                                                                                        |     | List-II |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|
| (P) | If the distance of the plane passing through the point $P(1,1,1)$ and perpendicular to the line $\frac{x-1}{3} = \frac{y-1}{0} = \frac{z-1}{4}$ from the origin is $\frac{p}{q}$ (where p and q are coprime), then the value of $(p - q)$ is                                                  | (1) | 1       |
| (Q) | The interval $[0,4]$ is divided into n equal sub-intervals by the points $x_0 = 0, x_1, x_2, \dots, x_{n-1}, x_n = 4$ . If $\delta x = x_i - x_{i-1}$ for $i = 1, 2, 3, \dots$ then the value of $\lim_{\delta x \rightarrow 0} \sum_{i=1}^{n-1} x_i \delta x$ is                             | (2) | 2       |
| (R) | Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying $xf(x) + (1 - x)f(-x) = x^2 + x + 1$ for any real numbers x. The greatest real number M for which $f(x) \geq M$ for all real numbers x, is equal to $\frac{p}{q}$ , where p and q are coprime. The value of $(q - p)$ is | (3) | 3       |
| (S) | The area of the trapezium ABCD with $AB \parallel CD$ , $AD \perp AB$ and $AB = 3CD$ is equal to 4. A circle inside the trapezium is tangent to all of its sides. If the radius of the circle is r then the value of $4r^2$ , is                                                              | (4) | 8       |

- (A)  $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$
- (B)  $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$
- (C)  $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$
- (D)  $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

8. Match the following :

|     | List-I                                                                                                                                                                                                                                         |     | List-II |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|
| (P) | Total number of values of $x$ , of the form $\frac{1}{n}$ , $n \in \mathbb{N}$ in the interval $\left[\frac{1}{15}, \frac{1}{10}\right]$ which satisfies $\{x\} + \{2x\} + \dots + \{12x\} = 78x$ where $\{ \}$ denotes fractional part of $x$ | (1) | 1       |
| (Q) | Let $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$ then the value of $\frac{2a_9}{a_{11}}$ is equal to                                                                                                                                       | (2) | 2       |
| (R) | Let $G$ be centroid of $\triangle ABC$ whose side length are $a, b, c$ . If $P$ is a point in the plane of $\triangle ABC$ such that $PA = 1$ , $PB = 2$ , $PC = 1$ & $PG = 1$ then the value of $\frac{a^2 + b^2 + c^2}{9}$ is equal to       | (3) | 3       |
| (S) | If a derivable function $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ satisfies $f(xy) = f(x) + f(y)$ $\forall x, y \in \mathbb{R}^+$ . If $f(16) = 3$ then $\frac{1}{f(2)} + \frac{1}{f(4)}$ is equal to                                          | (4) | 4       |

(A)  $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

(B)  $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$

(C)  $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$

(D)  $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

9. Match the following :

|     | List-I                                                                                                                                                                                                                                                                  |     | List-II |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|
| (P) | If the distance between two parallel tangents having slope $m$ drawn to the hyperbola $\frac{x^2}{9} - \frac{y^2}{49} = 1$ is 2, then $\frac{2}{5} m $ is equal to                                                                                                      | (1) | 5       |
| (Q) | If a variable line has its intercepts on the coordinate axes $e$ and $e'$ , where $\frac{e}{2}$ and $\frac{e'}{2}$ are the eccentricities of a hyperbola and its conjugate hyperbola, then the line always touches the circle $x^2 + y^2 = r^2$ , where $r$ is equal to | (2) | 4       |
| (R) | If the equation of lines touching both parabolas $y^2 - 4x = 0$ and $x^2 + 32y = 0$ is $ax - 2y + b = 0$ ( $a, b \in \mathbb{R}$ ), then $(a + b)$ is equal to                                                                                                          | (3) | 2       |
| (S) | If the mid-point of a chord of the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ is $M(0, 3)$ and the length of chord is $\frac{8p}{5}$ , then $p$ is equal to                                                                                                          | (4) | 1       |

(A)  $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

(B)  $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

(C)  $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

(D)  $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$



10. Match the following List-I with List-II

|     | List-I                                                                                                                                                                                                                                                                  |     | List-II |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|
| (P) | Let $f(x) = x^2 \sin \frac{\pi}{x} (x \neq 0)$ , then the number of points in $\left(\frac{1}{21}, 1\right)$ at which $f(x) = 0$ is                                                                                                                                     | (1) | 18      |
| (Q) | Let $f(x) = x^2 + px + q + 3$ , if $f(f(1)) = f(f(-2)) = 0$ where $f(1) \neq f(-2)$ and $g(x) = x^2 + (8p^2 - 18p)x - q + 2$ , then $2g(2)$ is                                                                                                                          | (2) | 20      |
| (R) | Let $f(x) = x^9 - 3x^8 - 53x^7 - 8x^6 - 8x^5 - 8x^4 - 8x^3 - 8x^2 - 8x + 9$ , then $f(9)$ will be equal to                                                                                                                                                              | (3) | 23      |
| (S) | Let $L = \lim_{x \rightarrow 0} \left(1 + e^{-\frac{1}{x^2}} \cdot \tan^{-1} \frac{1}{x^2} + x^2 \cdot e^{-\frac{1}{x^2}} \cdot \sin \frac{1}{x^4}\right)^{e^{\left(\frac{1}{x^2}\right)}}$ , then $[15 \log_e L]$ is (where $[.]$ represents greater integer function) | (4) | 27      |

The correct option is :

- (A)  $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$  (B)  $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$   
 (C)  $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$  (D)  $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$

11. Match the following List-I with List-II

|     | List-I                                                                                                                                                                                                                                                                                 |     | List-II        |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------|
| (P) | A continuous and differentiable function $f(x)$ satisfies functional equation $f(x+y) = \frac{f(x)f(y)}{f(x)+f(y)}$ ; $f(1) = 16$ , then $f(4)$ equals                                                                                                                                 | (1) | $\frac{7}{12}$ |
| (Q) | Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function satisfying $f\left(\frac{x+y}{5}\right) = \frac{6+f(x)+f(y)}{5} \forall x, y \in \mathbb{R}$ and $f'(2) = 2$ , then area bounded by relation $ y  = h(x)$ , $x = 2$ and $x = -2$ where $h(x) =  f(x) - 4 $ is | (2) | 4              |
| (R) | Let $f$ be a differentiable function satisfying the relation $f(x+2y) = f(x) + f(2y) + e^{x+2y}(x+2y) - xe^x - 2ye^{2y} + 4xy$ $\forall x, y \in \mathbb{R}$ and $f'(0) = 1$ , then the value of $\lim_{x \rightarrow 0} \frac{f(2x) - 4x^2 - 2x}{x^2}$ is                             | (3) | 2              |
| (S) | Let $f(x)$ is a differentiable function satisfying $f(y) - f(x) = \frac{x^x}{y^y} f\left(\frac{y^y}{x^x}\right) \forall x, y \in \mathbb{R}^+$ , if $f'(1) = 1$ , then area bounded between the curve $f(x)$ and $g(x) = 2x - 2x^2$ is                                                 | (4) | 8              |

The correct option is :

- (A)  $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$  (B)  $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$   
 (C)  $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$  (D)  $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$

12. Match the following List-I and List-II :

|     | List-I                                                                                                                                                                                                                                                                                                                                                                                       |     | List-II |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|
| (P) | If A and B are non-singular real matrices such that $\det(A) = 3$ and $A^{-1} + B^{-1} = (A + B)^{-1}$ , then $\det(B) =$                                                                                                                                                                                                                                                                    | (1) | 190     |
| (Q) | Let $A = \begin{bmatrix} a & x & p \\ y & q & b \\ r & c & z \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ ( $a, b, c, x, y, z, p, q, r \in \mathbb{N}$ ).<br>If $\text{tr}(AB + AB^3 + AB^5 + \dots + AB^{19}) = 210$ , then the number of ordered triplets (p,q,r) is ..... ( $\text{tr}(A)$ denotes sum of principal diagonal elements of A) | (2) | 3       |
| (R) | Consider $A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 4 & 5 \\ 4 & 0 & 0 \\ 5 & 0 & 0 \end{bmatrix}$ . If $C = [c_{ij}]_{3 \times 3} = A^{20} + B$ and $c_{22}c_{33} - c_{23}c_{32} = 2^m$ , then $m =$                                                                                                                                  | (3) | 5       |
| (S) | Consider three matrices $A = [a_{ij}]_{3 \times 5}$ , $B = [b_{ij}]_{5 \times 9}$ , and $C = A \times B = [c_{ij}]_{3 \times 9}$ .<br>If is given that $a_{ij} = \frac{i}{3j-2}$ and $b_{ij} = \frac{j}{3i+1}$ , then value of the element $c_{28}$ is                                                                                                                                       | (4) | 0       |

(A)  $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(B)  $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$

(C)  $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

(D)  $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$

SECTION-II : (Maximum Marks: 24)

**MATCHING LIST TYPE INTEGER QUESTION (ANSWER AS FOUR DIGIT INTEGER)**

- This section contains **SIX (06)** questions.
- The answer to each question is a four digit integer ranging from 1111 to 4444, both inclusive
- Each question having two matching lists. Choices for the correct combination of options form List-I (P), (Q), (R), (S) and List-II (1), (2), (3), (4)
- Given matching type questions in which List-I (PQRS) match with List-II (1, 2, 3, 4) and represent your answer as four digit integer for example correct match for  $P \rightarrow 2, Q \rightarrow 3, R \rightarrow 1, S \rightarrow 4$ . Then you have fill the answer in Section-II as 2314.
- Each matching in List-II may be/may not be correct & can be used more than one time for matching in List-I. For example your answer can be 2233, 1333, 1433, 4321 etc...
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +4 If ONLY the correct numerical value is entered.

*Zero Marks* : 0 In all other cases.

1. Suppose  $A \in \{a_1, a_2, a_3, a_4, a_5\}$  where  $a_1 < a_2 < a_3 < a_4 < a_5$  and  $B \in \{1, 2, 3, 4, \dots, k\}$  then, match List-I with List-II

|     | List-I                                                                                              |     | List-II |
|-----|-----------------------------------------------------------------------------------------------------|-----|---------|
| (P) | No. of injective functions in $f : A \rightarrow B$ such that $f(a_i) \neq i$ for $k = 6$           | (1) | 2002    |
| (Q) | No. of non-decreasing function in $f : A \rightarrow B$ for $k = 11$                                | (2) | 309     |
| (R) | No. of strictly increasing functions in $f : A \rightarrow B$ such that $f(a_i) \neq i$ for $k = 9$ | (3) | 3003    |
| (S) | No. of non-increasing function in $f : A \rightarrow B$ for $k = 10$                                | (4) | 56      |

2.

| List-I |                                                                                                        | List-II |                                            |
|--------|--------------------------------------------------------------------------------------------------------|---------|--------------------------------------------|
| (P)    | $\frac{1}{n} \left( \sum_{0 \leq i < j \leq n} (i+j) \binom{n}{i} \binom{n}{j} \right)$ is equal to    | (1)     | $n \cdot \binom{2n}{n}$                    |
| (Q)    | $\left( \sum_{0 \leq i < j \leq n} \left( \binom{n}{i}^2 + \binom{n}{j}^2 \right) \right)$ is equal to | (2)     | $\frac{1}{2} (2^{2n} - 2^n \binom{2n}{n})$ |
| (R)    | $\sum_{0 \leq i < j \leq n} (i+j) \left( \binom{n}{i} - \binom{n}{j} \right)^2$ is equal to            | (3)     | $[(n+1) 2^n \binom{2n}{n} - 2^{2n}] n$     |
| (S)    | $\sum_{0 \leq i < j \leq n} \binom{n}{i} \binom{n}{j}$ is equal to                                     | (4)     | $\sum_{r=0}^{n-1} 2^n C_r$                 |

3. There are 16 equally skilled players  $S_1, S_2, S_3, \dots, S_{16}$  playing a knockout tournament. They are divided into 8 pairs at random. From each pair a winner is decided on the basis of a game played between the two players of the pair, and they enter into the next round the tournament proceed in the similar way.

|     | List-I                                                                                                              |     | List-II        |
|-----|---------------------------------------------------------------------------------------------------------------------|-----|----------------|
| (P) | Probability that exactly one of $S_1$ and $S_2$ is among the 4 winners of 2 <sup>nd</sup> round is                  | (1) | $\frac{1}{30}$ |
| (Q) | The probability that $S_1$ wins the tournament given that $S_2$ enters in the semifinals is                         | (2) | $\frac{1}{20}$ |
| (R) | The probability that $S_1$ wins the tournament given that $S_2$ enters into final is                                | (3) | $\frac{7}{30}$ |
| (S) | The probability that $S_1$ is among the 4 winners of 2 <sup>nd</sup> round given that $S_2$ wins the first round is | (4) | $\frac{2}{5}$  |

4.

|     | List-I                                                                                                                                                                       |     | List-II        |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------|
| (P) | The least value of $8 z - 7  + 6 z - 5 $ ; $z \in \mathbb{C}$ is                                                                                                             | (1) | 12             |
| (Q) | The least value of $\left  \frac{z - 7 + 11i}{\sqrt{2}} \right  + \left  \frac{z + 5 - i}{\sqrt{2}} \right  + \left  \frac{z + 4}{\sqrt{2}} \right $ ; $z \in \mathbb{C}$ is | (2) | 13             |
| (R) | The the least value of $ z - 3 - 4i  +  z - i  +  z  +  z - 1 $ ; $z \in \mathbb{C}$ is                                                                                      | (3) | 10             |
| (S) | The least value of $ z - 3 ^2 +  z - 5 + 2i ^2 +  z - 1 + i ^2$ is                                                                                                           | (4) | $5 + \sqrt{2}$ |

5. A variable plane cuts the x-axis, y-axis and z-axis at the points A,B and C respectively such that the volume of the tetrahedron OABC remains constant equal 32 cubic unit and O is the origin of the coordinate system.

|     | List-I                                                             |     | List-II                         |
|-----|--------------------------------------------------------------------|-----|---------------------------------|
| (P) | The locus of the centroid of the tetrahedron is                    | (1) | $xyz = 24$                      |
| (Q) | The locus of the point equidistant from O,A,B and C is             | (2) | $(x^2 + y^2 + z^2)^3 = 192xyz$  |
| (R) | The locus of the foot of perpendicular from origin to the plane is | (3) | $xyz = 3$                       |
| (S) | If PA, PB and PC are mutually perpendicular then the locus of P is | (4) | $(x^2 + y^2 + z^2)^3 = 1536xyz$ |

6. Let  $P(x) = x^5 + x^2 + 1$  has zeros  $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$  and  $Q(x) = x^2 - 2$ , then match List-I with List-II :

| List-I |                                                        | List-II |     |
|--------|--------------------------------------------------------|---------|-----|
| (P)    | $\prod_{i=1}^5 Q(\alpha_i) =$                          | (1)     | 20  |
| (Q)    | $\sum_{i=1}^5 Q(\alpha_i) =$                           | (2)     | -23 |
| (R)    | $\sum_{1 \leq i < j \leq 5} Q(\alpha_i) Q(\alpha_j) =$ | (3)     | 40  |
| (S)    | $\sum_{i=1}^5 Q^2(\alpha_i) =$                         | (4)     | -10 |

---

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. ....

I have read all the instructions  
and shall abide by them.

\_\_\_\_\_  
Signature of the Candidate

I have verified the identity, name and Form  
number of the candidate, and that question  
paper and ORS codes are the same.

\_\_\_\_\_  
Signature of the Invigilator

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